

1.4 Importance of HCI

اچھا ایچ سی آئی پیچیدہ ٹیکنالوجی کو روزمرہ کا آسان اوزار بناتا ہے؛ برا ڈیزائن الجھن اور خطرہ پیدا کرتا ہے، جیسے...

CS XII

Chapter 1: Computer Systems (HCI)

Lecture 07 of 15

★ Golden

1.4 Importance of HCI ★ Golden

LEARN IT

HCI ensures that machines are approachable and usable, turning complex technology into everyday tools. **Good design** makes technology simple and enjoyable (touchscreens). **Poor design** causes confusion and risks (unclear buttons on medical devices).

This is why HCI is not optional decoration. A confusing JazzCash confirm button can send money to the wrong number. A confusing school-portal "Submit" can lock a student out of an online exam. Importance questions in the board exam expect the five implications below, each with a sentence of explanation.

Implication	Description
1. Enhance productivity	Well-designed interfaces help users work faster, make fewer mistakes, and achieve goals efficiently.
2. Promotes inclusivity	Accessibility features (screen readers, voice commands) empower marginalized groups and ensure equal use of technology for everyone.
3. Drives innovation	Emerging tech such as AR/VR, AI and IoT rely strictly on solid HCI principles for people to actually adopt and use them.
4. Economic impact	User-friendly products gain higher adoption rates, customer loyalty and market success, driving profits.
5. Ethical responsibility	Responsible design ensures technology respects user privacy, reduces bias and supports human well-being.

WORKED EXAMPLE

Worked example: good vs poor HCI, same task.

Task: send Rs 2,000 through JazzCash to a parent.

Good HCI: large "Send Money" button, amount shown in words and numbers, last 4 digits of the number repeated, fingerprint confirm, green "Successful" plus SMS. Result: fast, few mistakes, the user trusts the app (productivity + economic loyalty + ethics of clear consent).

Poor HCI: tiny grey buttons, no amount-in-words, no confirm screen, no tick. Result: wrong number, no idea if it worked, user deletes the app and goes back to a bank queue.

Medical contrast from the source: unclear buttons on a medical device are not just ugly — they are dangerous. That is why ethical responsibility sits in this table.

REMEMBER IT

P-I-I-E-E: Productivity, Inclusivity, Innovation, Economic impact, Ethics. Sentence: "Please Include Innovative, Ethical Earnings."

COMMON MISTAKE

Mistake: Writing only "HCI is important because it makes apps look nice."

Fix: Looks are a tiny part of UI. Importance answers must name productivity, inclusivity, innovation, money and ethics. Quote the medical-device warning if you have time.

اردو میں سمجھیں

اچھا ایچ سی آئی پیچیدہ ٹیکنالوجی کو روزمرہ کا آسان اوزار بناتا ہے؛ برا ڈیزائن الجھن اور خطرہ پیدا کرتا ہے، جیسے طبی آلے پر غیر واضح بٹن۔ پانچ مضمرات: پیداواری بڑھتی ہے، شمولیت آتی ہے، جدت اپنائی جاتی ہے، معیشت مضبوط ہوتی ہے، اور رازداری و انصاف کی اخلاقی ذمہ داری پوری ہوتی ہے۔

EXAM TIP

Typical wording: "Why is HCI important? Discuss its implications." (8 marks) Write a short opening (good design vs poor design) then five numbered points with one example each. Local examples that earn marks: ATM PIN privacy (ethics), screen reader on a school portal (inclusivity), JazzCash one-tap send (productivity), a bank's app beating a rival because it is easier (economic), AR classroom app that students actually use (innovation). Do not skip ethics — examiners look for privacy and well-being.

CHECK YOURSELF

1. Give one example of poor design mentioned in the source.
2. Which implication is about screen readers and equal use?
3. How does HCI create economic impact?

Answers: 1. Unclear buttons on medical devices. 2. Promotes inclusivity. 3. User-friendly products get higher adoption, loyalty and profits.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. HCI matters because:

- a. It turns complex machines into usable tools
- b. It makes CPUs clock faster by magic
- c. It replaces maths
- d. It deletes ethics

MCQ 2. Usability includes:

- a. Effectiveness, efficiency, satisfaction
- b. Only colour
- c. Only NAND count
- d. Only price

MCQ 3. Poor HCI costs:

- a. Errors, time, accidents, abandoned products
- b. Nothing
- c. Only printer ink
- d. Only font licences

MCQ 4. Good HCI can:

- a. Widen who can use the system (accessibility)
- b. Reduce RAM physically
- c. Change OSI to 2 layers
- d. Remove the need for passwords always

MCQ 5. This is ★: expect a question:

- a. Why HCI is important — 3–5 points with examples
- b. Trace a K-map
- c. Write bubble sort
- d. Normalise to 5NF

MCQ 6. A usable result portal:

- a. Students find the mark without a tutorial
- b. Hides the mark in 12 menus
- c. Uses 4pt text
- d. Needs a CS degree

Extra short questions

1. Three reasons HCI is important.
2. Link to a Pakistani e-service.
3. HCI vs pretty graphics.
4. One safety example.

Extra long question

1. Write a 10-mark essay: importance of HCI with five points, each with a local example.
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Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Fewer errors; faster tasks; more users included; safer devices; product success.
2. If the tax/passport site is confusing, people pay agents — HCI failure with social cost.
3. A pretty unusable app still fails; a plain clear form can succeed.
4. Car dashboards / medical pumps — confusion harms people.

Long

1. Learnability (first-year portal); efficiency (JazzCash); safety (clinic software); accessibility (screen reader); business (app store ratings). Short intro/conclusion.

Next lecture: 1.5 Accessibility Principles